Gene detectability shrinks as effect size falls

Oligodendrocytes · 100 donors, 250 cells/donor, 5742 reads/cell · DESeq2

|LFC| = 2.5

|LFC| = 1.5

|LFC| = 0.5

Raw pseudobulk baseMean (log scale)

10^{5} 10^{4} 10^{3} 10^{2} 10^{1} 10^{0} 10^{-1} 10^{-2}

baseMean 15 (approx. power floor)

not expressed in oligo (floor)

Intermediate filament (n=20)
Forkhead (n=49, 12 n.e.)
Cytokine (n=53)
Homeodomain (n=196, 93 n.e.)
GPCR (n=288)
Growth factor (n=47)
Signal molecule (n=120)
TM signal receptor (n=425)
bHLH (n=108, 20 n.e.)
Nuclear receptor (n=46)
bZIP (n=54, 3 n.e.)
Ion channel (n=179)
HMG/Sox (n=30, 4 n.e.)
C2H2 ZF (n=747, 73 n.e.)
Other TF (n=409, 32 n.e.)
RNA splicing (n=124)
Ribosomal (n=161)
RNA metabolism (n=322)
Chromatin regulator (n=218)
Ubiquitin ligase (n=359)
LoF-intolerant (pLI ≥ 0.9) (n=2956)

Undetectable (power ≤ 0.4)
Unsure (0.4–0.8)
Detectable (power > 0.8)
Not expressed in Oligo
TF family (Lambert DBD)
Non-TF class (PANTHER)
LoF-intolerant (pLI ≥ 0.9)